CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · NEUROLOGY PHASE 2,3 · STARTS 2015–2024

Tau: 4 molecules stopped early, all monoclonal antibodies

7 stopped trials, but 4 molecules — all monoclonal antibodies: Tilavonemab, Gosuranemab, Posdinemab and Semorinemab. 4 of the stops were the trial's own verdict and 3 followed a decision taken elsewhere. Against 9.6% for all neurology trials with a resolved drug target and 6.4% across all neurology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

70.0%
7 of 10 closed trials (95% CI 39.7%–89.2%)

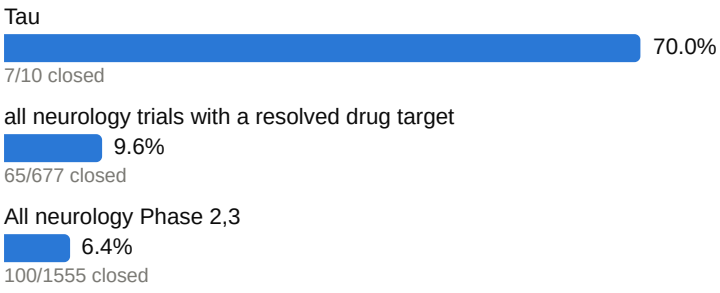
7 / 0 / 0
efficacy / safety / both (adds to 7)

100.0%
of 10 trials have closed · 0 still open or unresolved

40.0%
stopped for a biological reason within 3 years of starting (95% CI 17.3%–74.7%; 6 still at risk; 3.4% area-wide)

1 of 2 **Completed and missed the primary endpoint.** Of 3 completed trials in this segment, 2 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 1 missed and 0 met, 1 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 0 of the 3 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 5 distinct experimental drugs across 10 trials (antibody 10). Most tested: TILAVONEMAB (4), SEMORINEMAB (2), GOSURANEMAB (2), ZAGOTENEMAB (1), POSDINEMAB (1). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 7 stops came from 4 sponsor-asset programmes across 4 sponsors; the largest (AbbVie) contributed 3; removing that programme's trials from both sides leaves 66.7%. Screened alongside 27 other segments in this area, the rate stays unusual after a 10% false-discovery correction (q=0.000365, Benjamini-Yekutieli). Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2018–2020	3	5	60.0%	23.1%–88.2%

Where the stops sit

Efficacy / futility: 7.
Sponsors with most stops: AbbVie (3), Biogen (2), Johnson & Johnson (1), Roche (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04619420	2	Johnson & Johnson JNJ	POSDINEMAB	2025-10	Efficacy	Posdinemab did not achieve statistical significance in slowing clinical decline
NCT03712787	2	AbbVie ABBV	TILAVONEMAB	2021-09	Efficacy	Discontinued because of lack of efficacy in the parent study (Study M15-566; NCT02880956).
NCT03352557	2	Biogen BIIB	GOSURANEMAB	2021-08	Efficacy	The study (NCT03352557) was terminated based on lack of efficacy following the placebo-controlled period readout.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT03289143	2	Roche	SEMORINEMAB	2021-01	Efficacy	Study was terminated during Open Label Extension period because analyses of placebo-controlled Blinded portion of study did not show any evidence of clinical efficacy or modulation of accumulation of tau PET signal at any of the doses studied.
NCT03391765	2	AbbVie ABBV	TILAVONEMAB	2019-12	Efficacy	This study was prematurely discontinued because the program for progressive supranuclear palsy was discontinued due to lack of efficacy.
NCT02985879	2	AbbVie ABBV	TILAVONEMAB	2019-11	Efficacy	This study was prematurely discontinued because the program for progressive supranuclear palsy was discontinued due to lack of efficacy of study drug.

The 6 most recent of 7 stopped trials. Every molecule behind all 7 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 neurology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 45% of industry neurology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 100.0% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.